

Ishaan Goel

igoeldxb@gmail.com | +1 771-201-7576 | linkedin.com/in/igoelddc | github.com/igoelddc

Education

Illinois Institute of Technology	Aug 2022 – May 2027
Master of Artificial Intelligence	3.75/4.00
Bachelor of Science in Applied Mathematics	3.70/4.00
Bachelor of Science in Data Science	3.68/4.00

Experience

Institute for Pure and Applied Mathematics, UCLA	Los Angeles, CA
<i>Undergraduate Researcher, RIPS REU, sponsored by the Aerospace Corporation</i>	Jun 2026 – Aug 2026
- Developed an Extended Kalman Filter fusing GPS, radar, and computer vision measurements for airport surface localization, reducing RMSE by 59.7% over GPS alone and matching the accuracy of ASDE-X, a \$550M+ state-of-the-art, FAA surveillance system, at orders of magnitude lower cost. - Built a synthetic data pipeline using smoothing splines, Gauss-Markov noise modeling, and pinhole camera geometry to generate multi-sensor test trajectories from historical ADS-B data.	
Illinois Institute of Technology, Department of Applied Mathematics	Chicago, IL
<i>Research Assistant under Dr. Igor Cialenco, NSF-funded research</i>	Aug 2024 – Present
- Developing a multi-agent RL framework for stochastic groundwater markets under California's SGMA to analyze generalized Nash equilibrium behavior and policy outcomes under physical and regulatory constraints. - Implementing continuous-control RL policies with endogenous market clearing for groundwater consumption, banking, and rights-trading decisions over an infinite horizon.	
PeopleStrong Technologies Ltd.	Gurugram, India
<i>Artificial Intelligence Engineering Intern</i>	Jun – Jul 2025, Jul 2024
- Built a scalable RAG-based system for Automated Ticket Resolution, enabling document-grounded Q&A for 1,000+ employees and reducing index testing time by over 50x (weeks to hours). Deployed production-ready agentic AI systems and accelerated iteration cycles through automation and containerization. - Designed and deployed a scalable LLM-based chatbot with an agentic architecture integrating 5 RAG systems, validated via A/B testing and approved for company-wide rollout.	

Projects

Full-Stack Information Retrieval Engine <i>IR, scikit-learn, Flask, Scrapy, Python</i>	Oct 2025 – Dec 2025
Developed an information retrieval system using TF-IDF vector-space representations and cosine-similarity ranking, with dense semantic retrieval via Sentence-BERT and FAISS approximate nearest neighbors; evaluated lexical and semantic search using precision, recall, F1, and latency on a 200-document corpus.	
Distributed RL for Plastic Drift Simulation <i>RL, PyTorch, JAX, Python</i>	Aug 2025 – Nov 2025
Developed a distributed Monte Carlo RL framework that simulates ocean plastic drift and aggregates results from volunteer clients for large-scale model optimization, earning 3rd place in the Grainger Computing Innovation Prize.	

Relevant Coursework

Math: Probability, Linear Optimization, Stochastic Processes, Graph Theory, Monte Carlo Methods in Finance, SPDEs, Real Analysis, Abstract Algebra, Combinatorics, Dynamical Systems, Functional Analysis (audit)
AI/ML: Machine Learning, Statistical Learning, Natural Language Processing, Artificial Intelligence, Data Mining, Deep Learning, Information Retrieval, Algorithms, Database Organization, Big Data, Game Theory (audit)
In Progress: Social Graph Analysis & Graph Learning, Agentic AI for Campus Innovation

Leadership Roles

Vice President , SIAM Chapter at Illinois Institute of Technology	Dec 2022 – Present
Organize networking events for students across multiple disciplines related to mathematics. Connect students to research and industry opportunities in applied mathematics. Host professor- and student-led talks.	
President , Machine Learning at Illinois Institute of Technology (ML@IIT)	Jan 2023 – May 2025
Organized workshops and speaker events on AI/ML. Scaled organization from small group to 300+ active participants.	

Skills

Programming: Python, R, Julia, C++, Java, $\LaTeX$, HTML, CSS, SQL, Mathematica, MATLAB
Python Stack: PyTorch, scikit-learn, NumPy, pandas, matplotlib, NetworkX, PySpark
R Stack: tidyverse, data.table, glmnet, boot, gam, keras, caret, e1071, rpart, foreach, dbscan, pROC, Matrix, pls, class